

Question 2

Remark:

Strictly apply the definitions of pointwise, mean-square and uniform convergence. Pay close attention to the order in which limit or max/sup is applied in the mathematical definition.

(a) $y_N(t) = e^{-Nt} \cdot u(t)$ where $u(t)$ is the step function with $u(t) = 1$ at $t=0$.

(i) Pointwise convergence:

For $t < 0$: $y_N(t) = 0$

$\therefore$ The sequence $y_1(t), y_2(t), \dots$ is $0, 0, \dots$

$$\Rightarrow \lim_{N \rightarrow \infty} y_N(t) = 0.$$

For $t = 0$: $y_N(t) = 1$

$\therefore$ The sequence $y_1(t), y_2(t), \dots$ is $1, 1, \dots$

$$\Rightarrow \lim_{N \rightarrow \infty} y_N(t) = 1$$

For $t > 0$: $y_N(t) = e^{-Nt}$

$\therefore$ The sequence $y_1(t), y_2(t), \dots$ is

$$e^{-t}, e^{-2t}, e^{-3t}, \dots$$

$$\therefore \lim_{N \rightarrow \infty} y_N(t) = 0.$$

$$\therefore \text{choose } x(t) = \begin{cases} 0 & \text{if } t < 0 \\ 1 & \text{if } t = 0 \\ 0 & \text{if } t > 0 \end{cases}$$

— (1)

Then $y_N \rightarrow x$ pointwise.

(ii) Mean-Square Convergence.

Remember that we need to check if

$$\lim_{N \rightarrow \infty} \|y_N - x\|^2 = 0.$$

First find $\|y_N - x\|^2$,

choose $x(t)$ as in equation (1).

$$y_N(t) - x(t) = \begin{cases} 0 & \text{if } t \leq 0 \\ e^{-Nt} & \text{if } t > 0 \end{cases} \quad \rightarrow (2)$$

$$\begin{aligned} \|y_N - x\|^2 &= \int_0^{\infty} e^{-2Nt} dt = \left[\frac{e^{-2Nt}}{-2N} \right]_0^{\infty} \\ &= \frac{1}{2N} \end{aligned}$$

$$\therefore \lim_{N \rightarrow \infty} \|y_N - x\|^2 = \lim_{N \rightarrow \infty} \frac{1}{2N} = 0$$

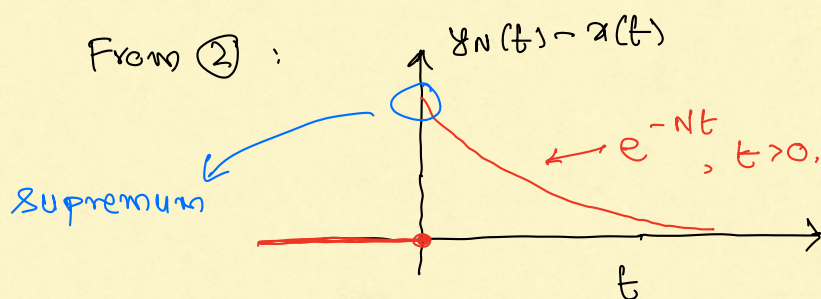
$\therefore y_N \rightarrow x$ uniformly.

(iii) Uniform convergence:

We need to check if $\sup_t |y_N(t) - x(t)| \rightarrow 0$
as $N \rightarrow \infty$.

So first find $\sup_t |y_N(t) - x(t)|$.

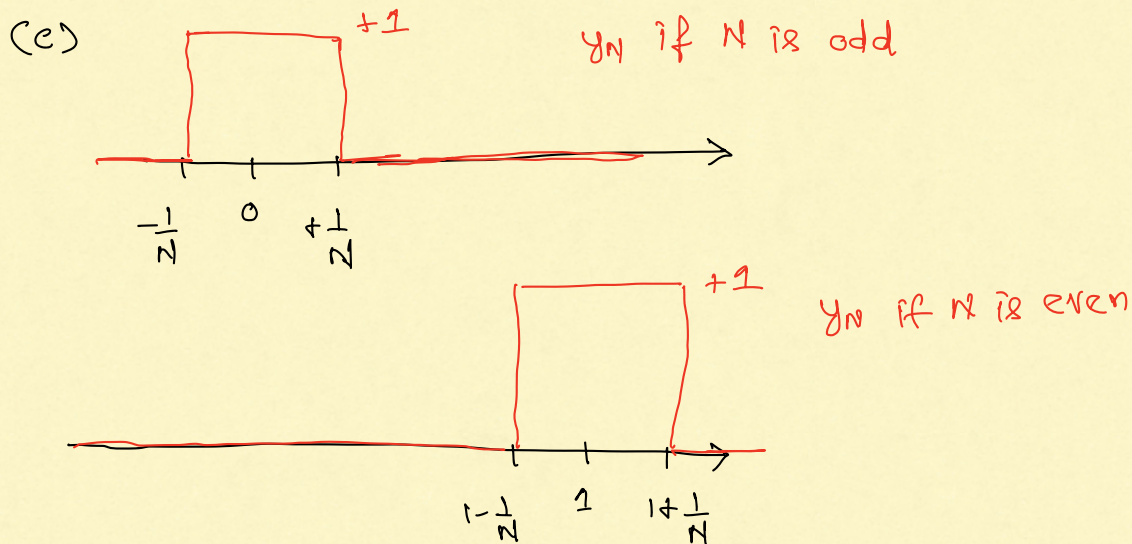
From (2):



$$\sup_t |y_N(t) - x(t)| = \sup_{t \geq 0} e^{-Nt} = 1.$$

$$\therefore \lim_{N \rightarrow \infty} \left[\sup_t |y_N(t) - x(t)| \right] = \lim_{N \rightarrow \infty} 1 = 1 \neq 0.$$

$\therefore y_N \not\rightarrow x$ uniformly.



(i) Pointwise convergence:

Pointwise convergence can happen only when

$\lim_{N \rightarrow \infty} y_N(t)$ exists for each and every t . $\rightarrow$ (S)

Non-trivial things are happening near $t=0$ and $t=1$.

So let us check this first.

What is the sequence $y_1(t), y_2(t), \dots$ at $t=0$?

This is: $1, 0, 1, 0, 1, 0 \dots$. This sequence does not have a limit.

Clearly property stated in (S) is not satisfied

$\Rightarrow y_N$ does not converge pointwise to any function.

Remark: For $t \neq 0, 1$: $\lim_{N \rightarrow \infty} y_N(t) = 0$. $\rightarrow$ (4)

But convergence fails at $t=0$ and $t=1$.

(ii) Mean-Square Convergence:

Based on (4), consider $x(t) = 0$ for all t .

First find $\|y_N - x\|^2$ and then do $\lim_{N \rightarrow \infty}$.

$\|y_N - x\|^2 = 2/N$ for both even & odd N .

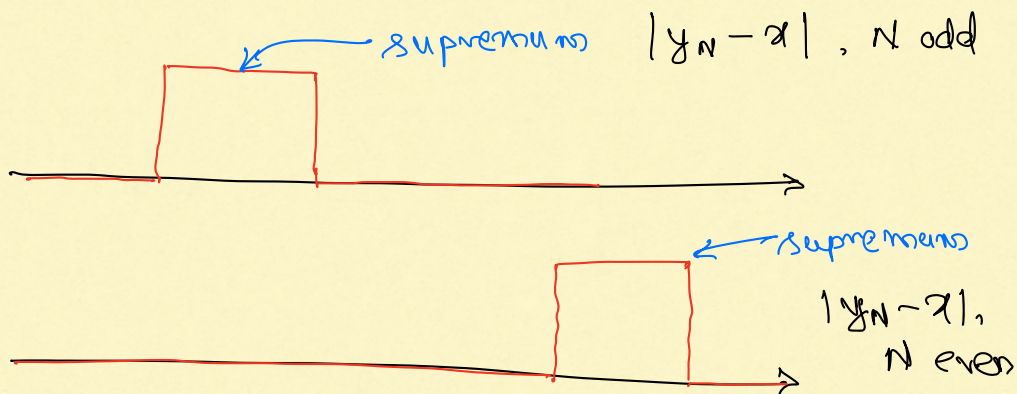
$$\lim_{N \rightarrow \infty} \|y_N - x\|^2 = \lim_{N \rightarrow \infty} \frac{2}{N} = 0.$$

$\therefore y_N \rightarrow x$ in the mean-square sense.

(iii) Since $y_N(t) \rightarrow 0$ for all $t \neq 0, 1$, any signal that could potentially be the limit of uniform convergence must have $x(t) = 0$ $\forall t \neq 0, 1$.

For such an $x(t)$:

$$\sup_t |y_N(t) - x(t)| = 1$$



$$\therefore \lim_{N \rightarrow \infty} \left[\sup_t |y_N(t) - x(t)| \right] = 1$$

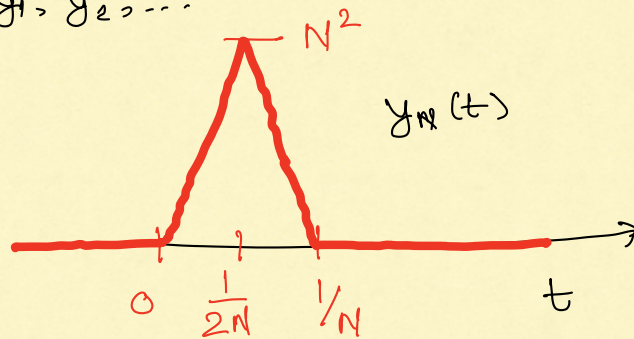
$\therefore y_N$ does not converge uniformly to any function.

Q-3

HINT:

consider the following sequence of functions

$y_1, y_2, \dots$



Proceed to show :

$$\lim_{N \rightarrow \infty} [y_N(t)] = 0 \quad \text{for all } t$$

$\therefore$ If $x(t) = 0$ $\forall t$ then

$y_N \rightarrow x$ pointwise

Also, check that $\|y_N - x\|^2 = 2 \times \int_0^{1/(2N)} (N^2 t)^2 dt$

$$= 2N^4 \left[\frac{t^3}{3} \right]_0^{1/(2N)}^{1/2N}$$

$$= \frac{2N^4}{3} \times \frac{1}{8N^3} = \frac{N}{12}$$

$$\therefore \|y_N - x\|^2 = N/12 \rightarrow \infty \text{ as } N \rightarrow \infty$$

$\therefore y_N \not\rightarrow x$ in the mean-square sense.

Q.5 HINT: We want f to have energy much smaller than f' .

Consider $f(t) = A \cdot e^{jn\omega t}$ with $\omega = \frac{2\pi}{T}$.

$$\|f\|^2 = |A|^2 \cdot T.$$

$$f'(t) = A \cdot jn\omega \cdot e^{jn\omega t}.$$

$$\|f'\|^2 = |A|^2 n^2 \omega^2 \cdot T$$

choosing n to be large will help us.

Now, given $\epsilon, \delta > 0$.

choose A such that $|A|^2 T \leq \epsilon$

$$\text{i.e. } |A| \leq \sqrt{\epsilon/T}.$$

After choosing such an A (with $A \neq 0$),

choose any integer n such that

$$|A|^2 \cdot n^2 \omega^2 T \geq \delta$$

$$\text{i.e. } n^2 \geq \frac{2\delta}{|A|^2 \omega^2 T}$$

$\therefore$ It is possible to have $\|f\|^2$ very small

and $\|f'\|^2$ very large simultaneously, even

$$\text{if } \int_0^T f(t) dt = 0 \quad \text{--- Fact 1}$$

Wirtinger's says: $\|f\|^2$ is large $\Rightarrow \|f'\|^2$ is large --- Fact 2

These two facts do not contradict each other

Converse to Wirtinger's? $\|f\|^2$ is small $\Rightarrow \|f'\|^2$ is small

$\rightarrow$ Not a fact, this is not true.

Fact 1 shows that converse of Fact 2 is not true.

Q.6 (b)

HINT:

choose an appropriate value of t such that

$$S_{2N-1}(t) = \sum_{n=-(2N-1)}^{2N-1} c_n e^{jn\omega t} = \sum_{k=1}^N \frac{1}{(2k-1)} \sin \left[\frac{2\pi(2k-1)}{3} \right].$$

we have pointwise convergence to $f(t)$ if we have bounded variation and f is continuous at this t . So check for these two properties.

Q.7

Let $f \xleftrightarrow{F.S.} c_n$ and $f^{(k)} \xleftrightarrow{F.S.} b_n$

$$b_n = \left(\frac{j2\pi n}{T} \right)^k c_n$$

$$|b_n|^2 = \left(\frac{4\pi^2 n^2}{T^2} \right)^k |c_n|^2.$$

$\|f^{(k)}\|^2$ is finite. By Parseval's relation:

$$\sum_{-\infty}^{+\infty} |b_n|^2 \cdot T < \infty \quad \text{i.e.} \quad \sum_{-\infty}^{+\infty} |b_n|^2 \text{ is finite.}$$

So $\sum_{-\infty}^{+\infty} |b_n|^2 = B$, where B is finite.

Then $|b_n|^2 \leq B$ for all n .

$$\Rightarrow |c_n|^2 \leq B \cdot \left(\frac{T^2}{4\pi^2} \right)^k \cdot \frac{1}{|n|^{2k}}.$$

↓
call this A .

Q. 8a : $\gamma = 1 + \frac{p}{q}$

$$2\gamma = 2 + \frac{2p}{q}$$

⋮

The decimal part of γ is p/q

"

2γ is $\langle 2p/q \rangle = \text{decimal part of } 2p/q$

⋮

"

$n\gamma$ is $\langle n \frac{p}{q} \rangle$.

⋮

"

$q\gamma$ is $\langle q \frac{p}{q} \rangle = \langle p \rangle = 0$.

"

$(q+1)\gamma$ is $\langle (q+1) \frac{p}{q} \rangle = p/q$.

(and then this repeats).

We only need to check

$$\langle p/q \rangle, \langle 2p/q \rangle, \dots, \langle (q-1)p/q \rangle, \langle qp/q \rangle \text{ --- (5)}$$

are distinct.

Distinctness: If $\langle mp/q \rangle = \langle np/q \rangle$, $m > n$
 $m, n \in \{1, \dots, q\}$

$$\Rightarrow \frac{mp}{q} = \frac{np}{q} + \text{same integer (say } a \text{)}$$

$$\Rightarrow a = \frac{(m-n)p}{q}$$

Since $m-n \in \{1, \dots, q-1\}$ and p & q have no common factors $\frac{(m-n)p}{q}$ is not an integer.

$\therefore$ This is a contradiction.

$\therefore$ the numbers in (5) are distinct.

Q-9

$$(a) \quad e^{j2\pi r} = \exp[j2\pi(\ell + \langle r \rangle)] \\ = \exp[j2\pi\ell] \cdot \exp[j2\pi\langle r \rangle]$$

$$e^{j2\pi r} = \exp[j2\pi\langle r \rangle].$$

$\langle kr \rangle = kr + \text{some integer}$
 $\downarrow$
decimal part of kr $\downarrow$
call this 'a'

$$g(\langle kr \rangle) = g(kr + a)$$

$$= \exp[j2\pi n(kr + a)]$$

$$= \exp[j2\pi nkr].$$

$$= a^k \quad \text{where } a = e^{j2\pi nr} \neq 1$$

$$\therefore \frac{1}{K} \sum_{k=1}^K g(\langle kr \rangle) = \frac{1}{K} \sum_{k=1}^K a^k = \frac{a^{K+1} - 1}{K(a-1)}$$

$$(b) \quad \left| \frac{1}{K} \sum_{k=1}^K g(\langle kr \rangle) \right| = \frac{1}{K} \times \frac{1}{|a-1|} \cdot |a^{K+1} - 1|$$

$$\leq \frac{1}{K|a-1|} \cdot (|a^{K+1}| + 1)$$

$\rightarrow$ triangle inequality

$$\leq \frac{1}{K|a-1|} \times 2 \quad (\text{since } |a^{K+1}| = 1)$$

$$\therefore \lim_{K \rightarrow \infty} \left| \frac{1}{K} \sum_1^K g(\langle kr \rangle) \right| = 0$$

$$\therefore \frac{1}{K} \sum_1^K g(\langle kr \rangle) \rightarrow 0 \text{ as } K \rightarrow \infty.$$

(c) For $n=0 \Rightarrow g(t) = 1$ for all t .

$$\therefore g(\langle kr \rangle) = 1$$

$$\therefore \frac{1}{K} \sum_1^K g(\langle kr \rangle) = 1.$$

(d) $S_N(t) = \sum_{-N}^{+N} c_n e^{j2\pi n t}$

$$\frac{1}{K} \sum_1^K S_N(\langle kr \rangle) = \frac{1}{K} \sum_{-N}^N \sum_1^K c_n e^{j2\pi n \langle kr \rangle}$$

$$= \sum_{-N}^{+N} c_n \cdot \frac{1}{K} \sum_1^K e^{j2\pi n \langle kr \rangle}$$

$$\lim_{K \rightarrow \infty} \frac{1}{K} \sum_1^K S_N(\langle kr \rangle) = \sum_{-N}^{+N} c_n \lim_{K \rightarrow \infty} \left[\frac{1}{K} \sum_1^K e^{j2\pi n \langle kr \rangle} \right]$$

$$= \sum_{-N}^{+N} c_n \cdot \delta_{n,0} = \begin{cases} 1 & \text{if } n=0 \\ 0 & \text{otherwise} \end{cases}$$

$$= c_0$$

$$= \frac{1}{T} \int_0^T f(t) dt, \text{ with } T=1, \omega=2\pi.$$